

Homework (Carolina Reaper)

- Q1.** A painter is creating a rhythm pattern for a music festival using $x \cdot y$ beats, where x is the number of drums and y is the number of beats per drum. If the pattern follows the linear equation $y = -2x + 10$, which point-slope form is equivalent?
- (A) $y - 2 = -2(x - 5)$
(B) $y - 1 = -2(x - 4)$
(C) $y - 3 = -2(x - 3)$
(D) $y - 4 = -2(x - 2)$
(E) $y - 5 = -2(x - 1)$
- Q2.** A DJ's playlist has a standard form equation of $2x + 5y = 20$, where x is the number of songs and y is the number of minutes per song. What is the slope-intercept form?
- (A) $y = -\frac{2}{5}x + 4$
(B) $y = -\frac{2}{5}x + 2$
(C) $y = -\frac{3}{5}x + 3$
(D) $y = -\frac{4}{5}x + 4$
(E) $y = -\frac{1}{5}x + 5$
- Q3.** A sound wave has a frequency of $5 \cdot 10^2$ Hz and follows the equation $y = 3x - 2$, where x is the time in seconds and y is the amplitude. Which standard form is equivalent?
- (A) $3x - y = 2$
(B) $3x - y = -2$
(C) $2x - 3y = 1$
(D) $2x + 3y = 6$
(E) $x - 3y = -2$
- Q4.** A music teacher wants to create a graph for a lesson plan. If the equation of a line is given in slope-intercept form as $y = 2x - 3$, which of the following is the point-slope form?
- (A) $y - 2 = 2(x - 1)$
(B) $y - 1 = 2(x - 2)$
(C) $y - 3 = 2(x - 0)$
(D) $y + 1 = 2(x + 2)$
(E) $y + 2 = 2(x + 1)$
- Q5.** The size of a musical instrument case follows a linear equation. If the slope-intercept form of the equation is $y = 4x - 2$, what is the y -intercept?
- (A) 2
(B) -2
(C) 4
(D) -4
(E) 0
- Q6.** A rhythm pattern has a linear equation of $x - 2y = 6$. If the equation is rewritten in slope-intercept form, what is the slope?
- (A) $-\frac{1}{2}$
(B) $\frac{1}{2}$
(C) -2
(D) 2
(E) $-\frac{3}{2}$
- Q7.** A DJ is planning a set with x number of songs, and each song has a duration of y minutes. The set follows the linear equation $y = 3x + 1$. If the DJ wants to play 5 songs, what is the total duration of the set?
- (A) 16 minutes
(B) 12 minutes
(C) 15 minutes
(D) 20 minutes
(E) 10 minutes
- Q8.** A sound engineer is setting up a sound system for a concert. If the linear equation for the sound waves is $y = -2x + 5$, what is the x -intercept?
- (A) 1
(B) 2
(C) 2.5
(D) $\frac{5}{2}$
(E) $-\frac{5}{2}$

Open Ended Questions (FRQ)

- Q9.** A painter is creating an art piece with a rhythm pattern. If the linear equation of the pattern is $y = 2x - 3$ and the painter wants to add a new element to the piece, what is the equation of the line that is perpendicular to the given line and passes through the point (1, 4)? Show all work and provide the equation in slope-intercept form.
- Q10.** A musician is planning a music festival with x number of stages and y number of bands per stage. The linear equation for the festival is $2x + 5y = 20$. Graph the equation and provide the x - and y -intercepts. Additionally, interpret the slope and y -intercept in the context of the music festival.

Chef's Tips

- Provide a ruler for graphing
- Allow the use of a calculator for calculations
- Have students show all work for free response questions